Component 01 brief explanation

This is the foundation and first component that starts the system. That is the email after it arrives in the inbox (general email arrival mechanism), after that the user selects **"Analyze this email"** through an add-on or extension. next that the component 01 analyze if the email of actually an AI-generated scam or not using it's methods (component 01's detection methods. Such as analyze uncertain signals and etc)

Complete Component 01 overview.

When user click the "Analyze this email" option for the selected email, next component 01 insert .eml file to its environment. After Whatever arrived (raw file, JSON, or text or .eml) is turned into one standard internal objects like subject, body, sender, headers, links, attachments. Then it extract the features that need to clearly identified the excatly is this one really AI genetated scam or not. So these are the filtering mechnaisms use for that. they are,

If selected, email passing all above filters to say accurately is the email originaly AI scam, then it must face these. They are,

Check	**What it checks**
assess scam-related risk	
SPF authentication	Whether the sending server is authorized to send email for the envelope-sender domain.
DKIM verification	Whether the email has a valid domain signature and the signed content remains intact.
DMARC alignment	Whether a passing SPF or DKIM identity aligns with the visible From domain.
Malicious sender-domain check	Whether the sender's domain appears in the locally cached threat blocklists.
Brand impersonation check	Whether the sender domain imitates a trusted organization through misspellings or lookalike characters.
Reply-To mismatch	Whether replies are directed to an address different from the sender, which may warrant investigation.
DNS/mail-record check - optional	Whether the sender domain has relevant DNS/mail records.

Social-engineering language	Whether the subject or body contains urgency, threats, secrecy, fake prizes, payment demands, or remote-access installation requests.
Malicious URL check	Whether email links match known-malicious URLs or domains in the cached feeds.
Suspicious URL structure	Whether links use shorteners, raw IP addresses, or disguised destinations.
Link-text mismatch	Whether the displayed destination differs from the actual link destination.
Attachment-risk check	Whether attachments have potentially dangerous types, macro-enabled formats, double extensions, or archive formats. This does not itself confirm malware.
DistilBERT scam classification	Whether the email text resembles scam patterns learned by the fine-tuned model.
assess possible AI writing	
Text predictability	Uses distilgpt2 to assess how predictable the wording is, as a supporting AI-writing signal.
Sentence-length variation	Measures variation in sentence lengths as a supporting writing-style feature.
Vocabulary repetition	Measures repeated words or phrases as a supporting AI-writing feature.
AI-writing classification	Uses the small classifier's trained features to estimate whether the text is AI-written.
determines whether AI analysis can run	
Minimum-text-length check	Determines whether the body contains enough words to run the AI-writing assessment.

Using together, they can support a **"**Suspected AI-assisted scam**".**

so here there is some conditionals,

There are short emails, or ones already judged legitimate, that skip above all these steps.

So after all this, generate the risk score. Here is the mechanism

$$z = b + w_m p_m + \sum_{i=1}^{n} w_i \, x_i$$

$$p_{\text{combined}} = \frac{1}{1 + e^{-z}}$$

Symbol definition.

p_m - DistilBERT's scam probability (assess scam-related risk – DistilBERT model)

x_i - A scam-related feature: malicious URL match, impersonation, credential request, authentication result, etc. (assess scam-related risk – without DistilBERT model)

w_m, w_i - Weights learned from labelled training data, not manually invented

b - Learned baseline value

w_m, w_i, b = numbers learned when training the combination model using those inputs and scam/legitimate labels.

$$\text{Scam Risk score} = 100 \times \text{Calibrate}(p_{\text{combined}})$$

When the risk score has not include the follwing lists,

- AI-writing probability.
- Minimum text length.
- Whether the AI detector ran.
- C3's steganography score; send that separately as image evidence.

After the risk score is calucluated then the particular email's (selected email's) ,

- Generate scam risk score
- Supporting evidence (explanabile/ resonable evidence)
- Anaysis status and missing checks (indicators)
- AI- authoriship assessment (possible AI involment)

Pass component 2 as the ouputs. Now the email detection system process is over. After that comes outputs as the inputs of the component 2 , which is "Human Centered Defense and Awareness System" is start.